Venkata Lakshmi Parimala Pasupuleti

Business Manager · Financial Analytics & Risk · Business Strategy · Quantitative Analysis · Pricing · Python · SQL · Tableau

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA — Open to McLean VA / NYC

PROFESSIONAL SUMMARY

Business Manager with 8+ years of financial analytics and risk modeling across large-scale consumer-facing platforms — managing end-to-end financial services functionalities including pricing strategy, credit verification, income analysis, and consumer regulatory compliance across 100K+ unit portfolios. **Drove people management and talent development** mentoring 5+ engineers and 600+ students. Built **quantitative analytical models achieving $R^2 = 0.977$** across 190+ countries; designed **hypothesis testing pipelines** on 1.6M+ financial records. Eliminated **80% of manual operational workload** through business strategy automation with zero compliance violations. Delivered strategic dashboards improving decision-making comprehension **40%** (Tableau, ggplot2). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 – Aug 2024

Senior Software Engineer — Financial Analytics · Business Strategy · Risk Modeling · Pricing · People Management

- Managed end-to-end **financial services functionalities** for property management and affordable housing: HAP allocation, **rent pricing strategy** (rent schedules, gross rent updates, AMI-based pricing tiers), tax credit calculations, HUD payments, rental subsidies, ledger management, and payment processing — analytical, **quantitative risk analysis** and **business strategy** ensuring compliance with federal and state consumer housing regulations across 100K+ unit portfolios; collaborated cross-functionally with compliance, operations, and product stakeholders.
- Automated **income eligibility and credit verification**, AMI threshold calculations, and recertification workflows across Section 8/HCV, LIHTC, and HUD compliance modules — **analytical hypothesis testing** on consumer income data; zero compliance violations; built production AI pipeline processing **1.6M financial records** (Python, PyTorch, OpenAI API) replacing manual Excel operations.
- Eliminated **80% of manual operational workload** through business strategy automation (Python/Bash pipelines); drove **people management and talent development** mentoring 5+ junior engineers across 300+ peer code reviews; achieved **100% PHI compliance** on AWS S3 migration — zero security incidents over 7+ years; optimized **200+ SQL queries** (30% performance gain) across 1M+ property records at 99.5% uptime.

George Mason University

Aug 2024 – Aug 2025

Research Associate — Quantitative Analytics · Hypothesis Testing · Strategic Stakeholder Reporting

- Achieved **$R^2 = 0.977$** regression accuracy across 190+ countries by building multi-source quantitative prediction models (Python, R) on WHO and UK Road Safety datasets — data collection, transformation, **hypothesis testing**, and validation pipeline; quantitative analysis driving research **strategy** recommendations for faculty stakeholders.
- Surfaced statistically significant anomalies manual review missed across 3 research pipelines by applying **Isolation Forest** — improved downstream analytical model reliability for cross-functional faculty teams.
- Partnered with cross-functional faculty stakeholders to translate analytical outputs into strategic dashboards (Tableau, ggplot2, Quarto) — improved decision-making comprehension **40%** across two cohorts; mentored **600+ students** across MIS 303 as Graduate Teaching Assistant.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer — Quantitative Systems · Risk Mitigation · Direct-to-Consumer Marketing Operations

- Parsed natural-language survey requests from **marketing, real estate, and commercial survey stakeholders** into structured flight plans using **quantitative photogrammetry math** — enabling **direct-to-consumer marketing** and commercial aerial survey operations; every compliance check functions as a **hypothesis test** grounded in FAA Part 107 regulation.
- Engineered validation guardrails enforcing FAA altitude limits, battery budgets, and no-fly-zone checks — **risk mitigation** blocking non-compliant missions before export across marketing and commercial survey teams.

PROJECTS

CrashClarity: Quantitative Risk Analytics on 104K Records with Self-Service Dashboard

Python · R · Scikit-learn · Random Forest · Logistic Regression · Hypothesis Testing · Feature Engineering · Tableau · GitHub Pages — Live Dashboard

- Mined 104K+ UK accident records — built Random Forest + Logistic Regression ensemble with hypothesis testing and feature engineering (weather, road, time) achieving **85%+ accuracy**; deployed interactive self-service dashboard translating quantitative model outputs into risk-severity insights for non-technical stakeholders.

Lease: Compliance Audit & Credit Verification Pipeline — 93.6% Precision

Python · LangGraph · Scikit-learn · SQLite · Deterministic Rules Engine · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an end-to-end compliance audit and credit verification pipeline across 1,240 consumer documents — feature extraction, deterministic rule-based classification, and audit engine with full data lineage; achieved **93.6% precision, 87.9% F1** across 10 compliance rules; every determination traceable and audit-ready.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Quantitative modeling, hypothesis testing, risk analytics, business strategy, financial data analytics, data visualization

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Business Analytics & Strategy: Quantitative Analysis, Financial Services Analytics, Risk Modeling, Hypothesis Testing, Pricing Strategy, Business Strategy, Credit Verification, Consumer Compliance, People Management, Talent Development, Cross-Functional Leadership, Stakeholder Reporting, Direct-to-Consumer Analytics, AMI Threshold Modeling, LIHTC/HCV/HUD Regulatory Workflows

Analytics & Modeling: Python (expert), R, SQL (expert — 200+ production optimizations, execution plan tuning, indexing, partitioning), PL/SQL; Statistical Modeling, Regression Analysis, Random Forest, Logistic Regression, XGBoost, Isolation Forest, Feature Engineering, Scikit-learn, Pandas, NumPy; Tableau, Power BI, ggplot2, Quarto, matplotlib

Cloud & Data Infrastructure: AWS (S3, EC2, RDS, Glue), Databricks, ETL/ELT Pipelines, PostgreSQL, MySQL, Docker, CI/CD (GitHub Actions), Salesforce (CRM integrations, reporting automations), Git, Linux, Agile/Scrum (120+ sprints)